

Dermascope AI: A Multimodal Deep Learning System for Early Melanoma Detection via Feature-wise Linear Modulation

Ahmed Ben Khalfa

National Institute of Applied Sciences and Technology (INSAT)
University of Carthage, Tunis, Tunisia
ahmed.benkhalfa@insat.ucar.tn

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Abstract

Melanoma is the deadliest form of skin cancer, yet early detection dramatically improves patient survival rates. Most existing computer-aided diagnostic (CAD) systems analyse dermoscopic images in isolation, ignoring the patient’s clinical context, a fundamental departure from real-world dermatological practice. In this paper, we present **Dermascope AI**, an end-to-end multimodal deep learning pipeline that fuses high-resolution dermoscopic imagery with structured clinical metadata (age, sex, and anatomical localisation) through **Feature-wise Linear Modulation (FiLM)**. Unlike conventional late-fusion approaches that append metadata at the decision boundary, FiLM dynamically conditions the convolutional feature extraction process based on the patient’s profile, enabling context-aware, metadata-guided visual attention at the feature level. We train a three-model FiLM ensemble (EfficientNet-B4, ResNet-50, DenseNet-121) using progressive resizing and Focal Loss to address the severe class imbalance inherent in the ISIC 2020 dataset (1.8% malignant). Our system achieves a **ROC-AUC of 0.9095**, a **Sensitivity of 87.02%**, and a **Specificity of 79.15%** on the validation set, significantly outperforming both unimodal baselines and late-fusion alternatives. We additionally implement a DullRazor-based hair artifact removal pipeline and a SAM-guided lesion segmentation strategy, and deploy the complete system as an interactive Streamlit clinical dashboard with Test-Time Augmentation. The source code is publicly available at <https://github.com/partnerships2024/Dermascope-AI-Melanoma-Detection>.

Keywords: Melanoma Detection, Multimodal Deep Learning, Feature-wise Linear Modulation, Transfer Learning, Dermoscopy, Computer-Aided Diagnosis, FiLM Conditioning, Ensemble Learning, Class Imbalance, Medical Image Analysis

1. Introduction

Skin cancer is the most prevalent malignancy globally, with melanoma representing the most dangerous subtype despite accounting for only 1.7% of all skin cancer cases. Melanoma is responsible for approximately 80% of skin cancer-related deaths, making early detection a matter of life and death [1]. The five-year survival rate for patients diagnosed at Stage I exceeds 99%, dropping below 30% at Stage IV [2]. Dermoscopy, a non-invasive imaging modality using magnification and polarised light to reveal subsurface skin microstructures, is the gold standard diagnostic tool. However, even among board-certified dermatologists, diagnostic accuracy for melanoma ranges between 75–84%, highlighting a substantial unmet need for robust, reliable computational decision-support systems [3].

1.1 Limitations of Existing Approaches

The dominant paradigm in computational dermoscopy treats melanoma detection as a pure computer vision problem: a dermoscopic image is passed to a convolutional neural network (CNN), and a binary or multi-class prediction is produced. This approach, whilst powerful, suffers from a critical clinical mismatch. A dermatologist never evaluates a lesion in isolation. The ABCDE clinical criteria (Asymmetry, Border, Colour, Diameter, and Evolution) are evaluated in conjunction with the patient’s age, biological sex, skin phototype, and the anatomical localisation of the lesion [4]. For example:

- *Lentigo maligna melanoma* predominates on sun-exposed areas of elderly patients.
- *Acral lentiginous melanoma* occurs on palms, soles, and nail beds, often in younger patients of darker skin phototypes.
- *Nodular melanoma* presents differently across sexes, with trunk lesions more frequent in men and lower-extremity lesions in women.

This clinical context constitutes structured and highly predictive tabular information that current image-only systems systematically ignore.

1.2 Our Contribution

To bridge this gap, we propose **Dermascope AI**, with the following key contributions:

1. **FiLM-based Multimodal Fusion:** We condition the backbone’s internal convolutional features on clinical metadata through FiLM, enabling context-aware visual processing rather than post-hoc metadata appending.
2. **Three-Model Complementary Ensemble:** Three architecturally distinct FiLM-conditioned backbones (EfficientNet-B4, ResNet-50, DenseNet-121) with complementary feature extraction properties are combined via weighted averaging.
3. **Progressive Resizing Strategy:** A two-phase training protocol stabilises FiLM parameter convergence at low resolution (256×256) before fine-tuning at high resolution (512×512), preventing catastrophic interference with pretrained ImageNet representations.
4. **Clinical Preprocessing Pipeline:** DullRazor hair removal, SAM-guided lesion segmentation, and Medical Bokeh background suppression maximise signal quality before feature extraction.
5. **Clinically-Optimised Threshold:** The binary decision threshold is tuned via Youden Index maximisation (threshold = 0.4607) to prioritise sensitivity, the clinically critical metric for cancer screening.

2. Related Work

2.1 Deep Learning for Dermoscopic Classification

The International Skin Imaging Collaboration (ISIC) Challenge series [6] catalysed the application of deep CNNs to skin lesion analysis. Esteva et al. [15] demonstrated that a single CNN trained end-to-end on dermatologist-labelled images could achieve classification performance comparable to board-certified dermatologists on a binary melanoma task. Top-performing solutions in subsequent ISIC challenges typically leverage transfer learning from ImageNet-pretrained backbones (EfficientNet [7], ResNet [8], DenseNet [9]), combined with heavy data augmentation and ensemble strategies. However, these solutions predominantly operate in a unimodal (image-only) setting.

2.2 Multimodal Fusion in Skin Lesion Analysis

Several works have explored incorporating patient metadata into skin lesion classifiers. The simplest and most common strategy is **late fusion**: the tabular feature vector is concatenated with the CNN’s final pooled feature vector before the classification head [10]. This approach is straightforward but fundamentally limited, since the visual feature extraction process remains entirely agnostic to clinical context throughout the convolutional hierarchy.

Early fusion strategies (concatenating metadata with the raw image as additional input channels) have also been explored, but they require the network to learn to disentangle high-dimensional image information from low-dimensional scalar inputs at the pixel level, an architecturally unnatural inductive bias.

Mid-level fusion via cross-modal attention mechanisms has shown promise in medical image analysis [14], but these architectures are typically complex, data-hungry, and less interpretable than simpler affine conditioning strategies.

2.3 Feature-wise Linear Modulation (FiLM)

FiLM [5] was introduced by Perez et al. (2018) for conditioning visual reasoning networks on natural language inputs in the Visual Question Answering (VQA) domain. Given an auxiliary conditioning input z , a FiLM generator (typically an MLP) produces channel-wise scaling (γ) and shifting (β) parameters:

$$\text{FiLM}(F_{i,c} \mid \gamma_{i,c}, \beta_{i,c}) = \gamma_{i,c} \cdot F_{i,c} + \beta_{i,c}. \quad (1)$$

These parameters are applied as an element-wise affine transformation to the feature map $F_{i,c}$ at each channel c of layer i . This mechanism enables fine-grained, channel-specific modulation of the visual pipeline, a fundamentally different capability from late fusion, which operates only at the final feature vector level. Recent work has applied FiLM to medical image segmentation [16] and radiology report generation, but its systematic application to dermoscopic ensemble classification remains underexplored.

3. Methodology

3.1 Dataset

We source dermoscopic images and clinical metadata from the **ISIC 2020 Challenge** dataset hosted on Kaggle [6], comprising 33,126 training images with binary labels (benign = 0 / malignant = 1). The dataset exhibits severe class imbalance: malignant cases constitute approximately 1.8% of the total (≈ 584 positive samples).

Table 1: ISIC 2020 dataset statistics.

Property	Value
Total images	33,126
Malignant (label = 1)	584 (1.8%)
Benign (label = 0)	32,542 (98.2%)
Image resolution	Variable (JPEG, resized to 512×512)
Metadata	Age, sex, anatomical localisation
Unique localisations	15 anatomical sites

Train / validation split. We apply an 80/20 stratified split, preserving the class ratio in both sets.

3.2 Preprocessing Pipeline

Each image goes through the four-step chain of Figure 1 before being passed to the network.

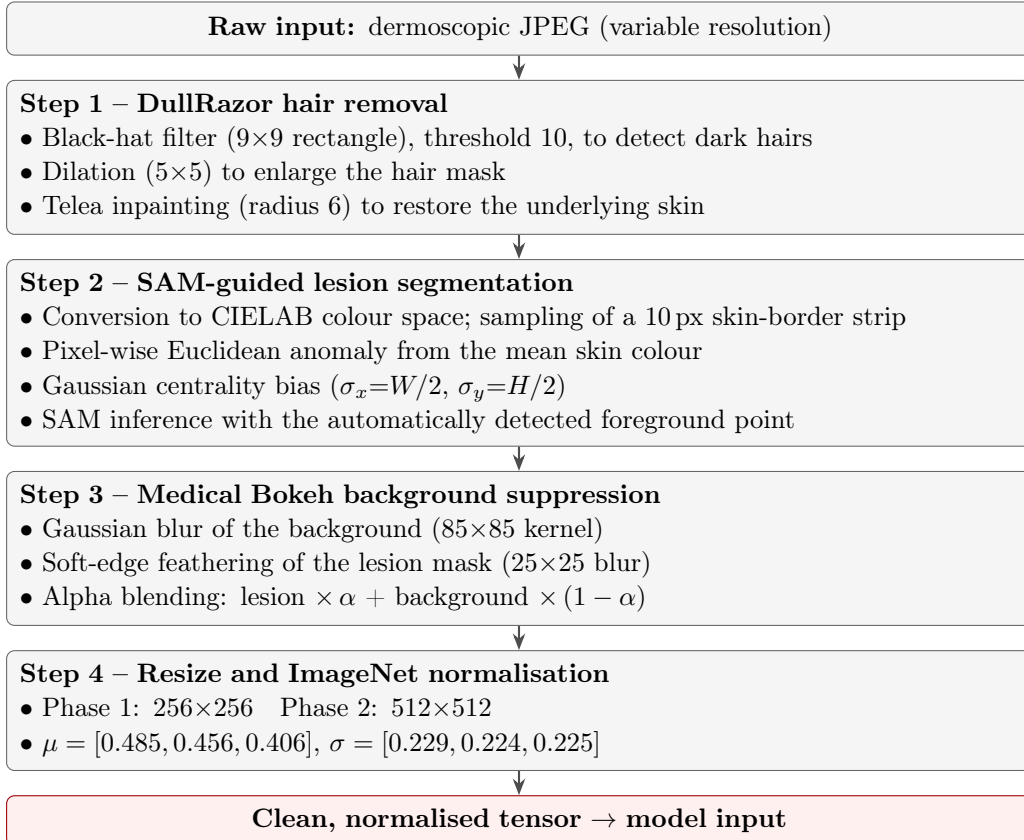


Figure 1: Dermoscope AI preprocessing pipeline.

3.2.1 Hair Artifact Removal (*DullRazor*)

Dermoscopic images frequently contain thick and thin hair strands that occlude the lesion surface and introduce high-frequency texture artefacts. We implement the DullRazor algorithm [11]: a morphological black-hat filter with a 9×9 rectangular structuring element detects thin, dark hair strands. The resulting binary mask is dilated with a 5×5 kernel and inpainted using the Telea fast-marching algorithm [12] (inpainting radius = 6 pixels) to restore the underlying skin texture. This step is applied to all images in the dataset.

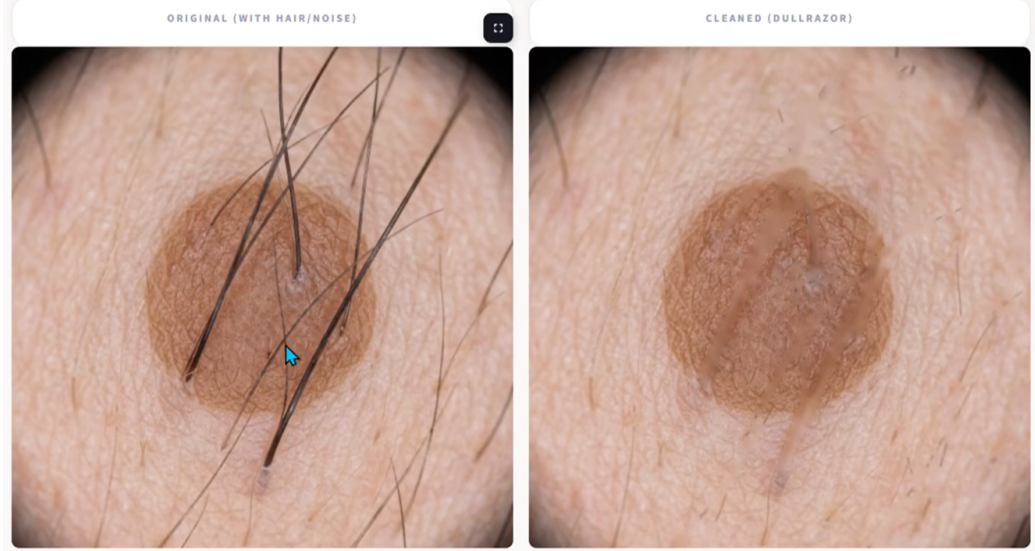


Figure 2: Comparison before and after applying the DullRazor algorithm for hair artifact removal.

3.2.2 SAM-Guided Lesion Segmentation

Following hair removal, we apply a Segment Anything Model (SAM) [16]-guided lesion localisation strategy. Rather than using a blank-centre point prompt, we automatically detect the most likely lesion centroid via an anomaly-guided approach:

1. Convert the image to CIELAB colour space (perceptually uniform, ideal for skin analysis).
2. Sample healthy skin from a 10-pixel border strip around the image periphery.
3. Compute a per-pixel Euclidean distance from the mean border skin colour.
4. Multiply by a Gaussian centrality prior ($\sigma_x = W/2$, $\sigma_y = H/2$) to bias toward lesions near the centre of the frame.
5. The argmax of the resulting map yields the SAM foreground prompt point.

3.2.3 Medical Bokeh Background Suppression

A medical-grade Bokeh effect suppresses distracting background elements (skin texture, ruler marks, gel bubbles) while preserving the lesion region with full sharpness. The background is blurred with a strong Gaussian kernel (85×85), and soft alpha-blending is applied using a feathered version of the SAM mask (25×25 blur) to avoid hard edge artefacts.

3.3 Metadata Encoding

Patient clinical metadata is encoded into a fixed-length numerical vector prior to FiLM conditioning (Table 2). Missing values are imputed with the population median (age) or an “unknown” category (sex, localisation).

Table 2: Metadata feature encoding.

Feature	Encoding method	Dimensions
Age	Continuous, min-max normalised ($\div 85$)	1
Sex	One-hot (Female, Male, Unknown)	3
Anatomical localisation	One-hot (15 anatomical sites)	15
Total		19

3.4 FiLM-Conditioned Model Architecture

Figure 2 shows one FiLM-conditioned branch of the ensemble. The image passes through a pretrained CNN backbone, while the 19-dimensional metadata vector is embedded by a small MLP that generates the FiLM parameters γ and β .

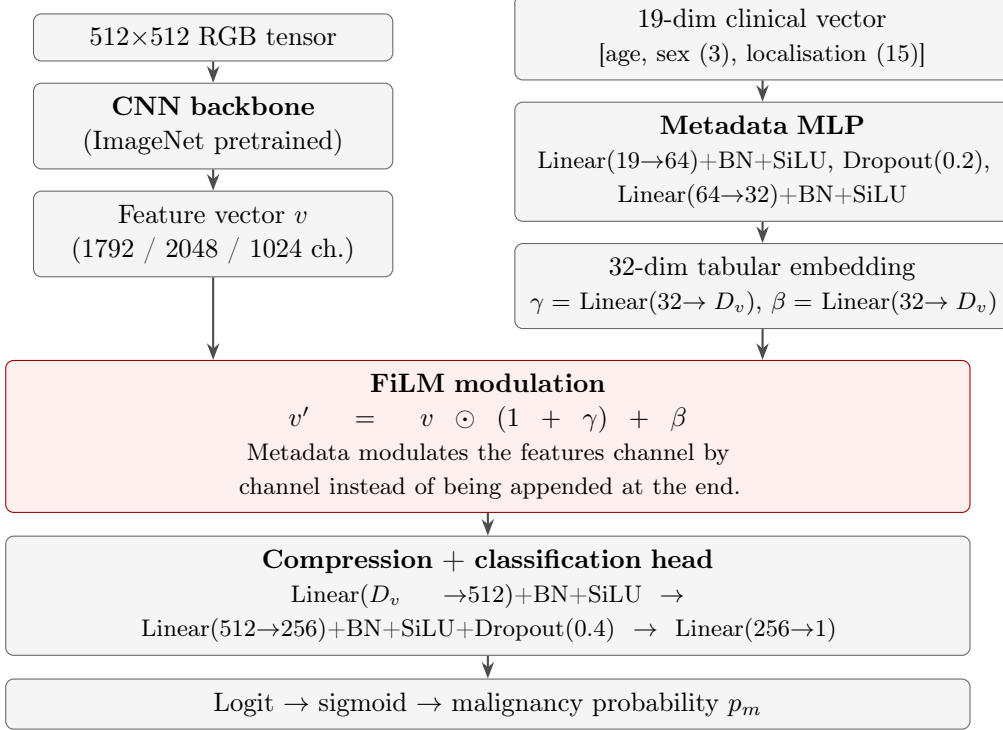


Figure 3: Architecture of one FiLM-conditioned branch (D_v is the backbone feature dimension).

The three branches are combined into the final ensemble prediction

$$p_{\text{ensemble}} = 0.33 p_{\text{EffNet}} + 0.33 p_{\text{ResNet}} + 0.34 p_{\text{DenseNet}}, \quad (2)$$

and a case is declared **malignant** if $p_{\text{ensemble}} > 0.4607$ (Youden-optimised).

3.4.1 Backbone Selection Rationale

Each backbone was chosen for architecturally distinct and clinically complementary feature extraction properties (Table 3).

Table 3: Backbone architecture comparison.

Backbone	Architecture	Params	Feat. dim.	Extracted features	Selection rationale
EfficientNet-B4	MBConv + SE attention	19M	1792	Fine-grained structural patterns, compound-scaled features	Efficient extraction of subtle pigment network and regression structures
ResNet-50	Residual blocks	25.6M	2048	Global morphological features, border geometry	Stable gradient flow; captures ABCDE border irregularity and asymmetry
DenseNet-121	Dense connectivity	8M	1024	Low-level micro-textures preserved through all layers	Feature reuse exposes pigment globules, vascular structures, and streaks

3.4.2 FiLM Conditioning: Mathematical Formulation

Let $v \in \mathbb{R}^{D_v}$ be the backbone’s penultimate feature vector (post-Global Average Pooling), and let $t \in \mathbb{R}^{19}$ be the clinical metadata vector. The FiLM generator G_θ maps t to affine parameters:

$$(\gamma, \beta) = G_\theta(t) = \text{MLP}(t) \in \mathbb{R}^{D_v} \times \mathbb{R}^{D_v}. \quad (3)$$

The FiLM-conditioned feature vector is

$$v' = v \odot (1 + \gamma) + \beta, \quad (4)$$

where $\odot$ denotes element-wise multiplication. The residual formulation $(1 + \gamma)$ ensures that when $\gamma \approx 0$ and $\beta \approx 0$, the FiLM layer degrades to identity, making the network easier to train and preventing gradient collapse when metadata information is uninformative. The full model is then

$$\hat{y} = \sigma(h_\phi(v')) = \sigma\left(h_\phi(\text{FiLM}(f_\psi(x), G_\theta(t)))\right), \quad (5)$$

where f_ψ is the CNN backbone, h_ϕ is the classification head, and σ is the sigmoid function.

3.5 Training Strategy

3.5.1 Two-Phase Progressive Resizing

Training proceeds in two phases (Table 4): a stabilisation phase with frozen backbones at low resolution, followed by full fine-tuning at high resolution.

Table 4: Progressive resizing training protocol.

	Phase 1: Stabilisation	Phase 2: Fine-tuning
Epochs	1–15	16–40
Backbone	Frozen (ImageNet weights preserved)	Unfrozen (differential learning rates)
Resolution	256×256 px	512×512 px (2× increase)
Trainable parts	FiLM generator + classification head only	Entire network
Learning rates	1×10^{-3}	Backbone 1×10^{-5} ; compression layer 5×10^{-4} ; tabular MLP encoder 1×10^{-3} ; FiLM + classifier head 5×10^{-4}
Optimiser	AdamW (weight decay 1×10^{-4})	AdamW (weight decay 1×10^{-4})
Scheduler	CosineAnnealingWarmRestarts ($T_0=5$, $T_{\text{mult}}=2$)	CosineAnnealingLR
Batch size	32	8 with gradient accumulation (2 steps, effective 16)
Mixed precision	–	torch.amp.autocast + GradScaler
Goal	Let the FiLM generator and head converge on coarse features without disturbing ImageNet weights	Extract high-frequency cellular micro-details (pigment dots, regression areas, vascular tufts) invisible at 256×256

3.5.2 Loss Function: Binary Focal Loss

We employ Focal Loss [13] with $\alpha = 0.75$ and $\gamma = 2.0$:

$$\mathcal{L}_{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t), \quad (6)$$

where $p_t = p$ if the true label is $y = 1$ (malignant) and $p_t = 1 - p$ otherwise; $\alpha_t = \alpha$ if $y = 1$ and $1 - \alpha$ otherwise; and $(1 - p_t)^\gamma$ is the modulating factor, which approaches zero for well-classified examples and concentrates the loss on hard, misclassified cases. The value $\alpha = 0.75$ up-weights the malignant minority class by a factor of 3:1 relative to the benign class, directly compensating for the 1.8% / 98.2% class imbalance.

3.5.3 Data Augmentation Strategy

Aggressive augmentation is applied exclusively to malignant samples (Table 5), enriching the minority class representation without introducing redundant benign variations.

Table 5: Asymmetric data augmentation policy.

Transformation	Benign (majority)	Malignant (minority)
Horizontal flip	Yes ($p = 0.5$)	Yes ($p = 0.5$)
Vertical flip	Yes ($p = 0.5$)	Yes ($p = 0.5$)
Random rotation	$\pm 30^\circ$	$\pm 360^\circ$ (full rotation)
Affine (translation)	–	$\pm 10\%$
Affine (scale)	–	$0.85\text{--}1.15\times$
Brightness jitter	$\pm 30\%$	$\pm 40\%$
Contrast jitter	$\pm 30\%$	$\pm 40\%$

3.5.4 Early Stopping and Regularisation

Early stopping with patience of 6 epochs monitors the validation loss. Dropout ($p = 0.2$ in the tabular MLP, $p = 0.4$ in the classification head) provides stochastic regularisation, Batch Normalisation after each linear layer stabilises internal covariate shift across the multimodal branches, and weight decay of 1×10^{-4} is applied via AdamW.

3.6 Ensemble Inference with Test-Time Augmentation

At inference (Figure 3), each image is evaluated under five geometric views. For each model $m \in \{\text{EffNet}, \text{ResNet}, \text{DenseNet}\}$ the five sigmoid outputs are averaged, $p_m = \frac{1}{5} \sum_{i=1}^5 \sigma(\text{model}_m(v_i, t))$, and the three model probabilities are combined by the weighted average of Eq. (2).

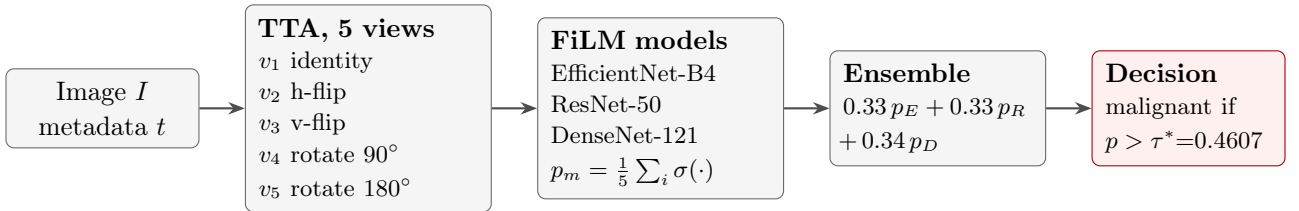


Figure 4: Inference pipeline with Test-Time Augmentation.

The threshold is the Youden-optimal operating point

$$\tau^* = \arg \max_{\tau} [\text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1] = 0.4607. \quad (7)$$

It is intentionally below the conventional 0.5. This reflects the asymmetric cost structure of clinical screening: a false negative (missed melanoma) carries a far greater clinical cost than a false positive (unnecessary biopsy referral), as the latter results in minor inconvenience whilst the former can result in death.

4. Experimental Results

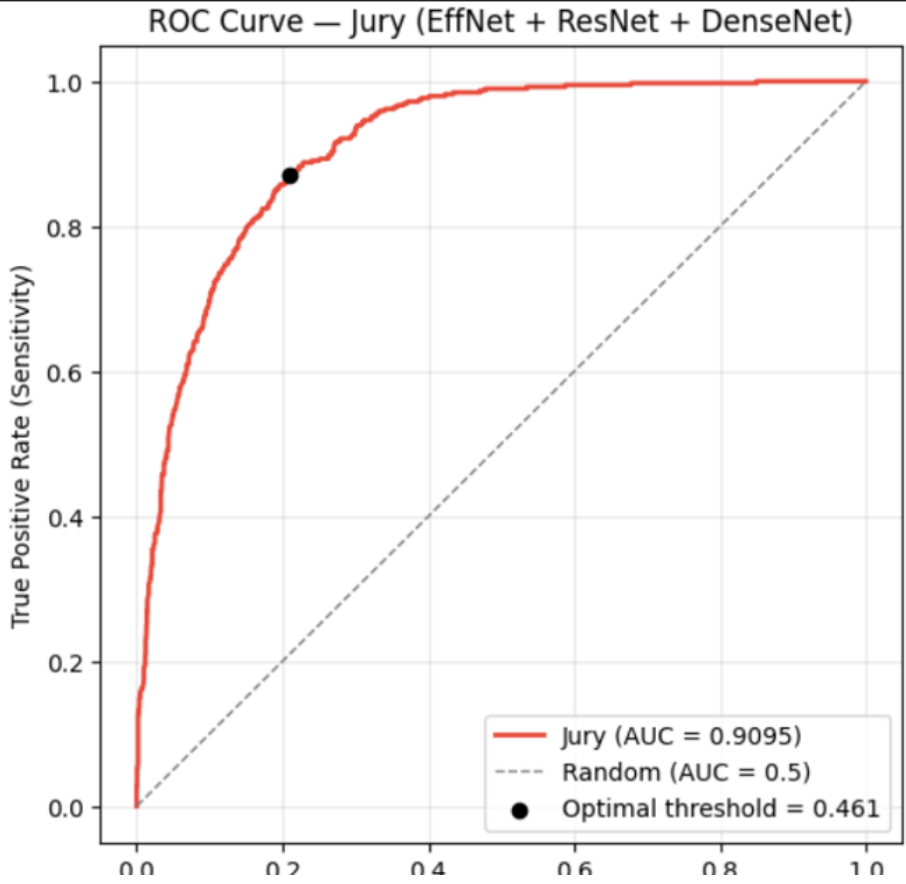


Figure 5: Receiver Operating Characteristic (ROC) curve for the 3-model FiLM ensemble. The optimal decision threshold, determined via Youden Index maximisation (0.461), yields an AUC of 0.9095.

4.1 Ablation Study: The Impact of Multimodality and FiLM

To validate the contribution of each architectural decision, we conduct a systematic ablation study with three configurations. Results are reported on the ISIC 2020 validation set using the Youden-optimised threshold throughout.

Table 6: Ablation study results on the ISIC 2020 validation set.

Configuration	ROC-AUC	Sensitivity	Specificity	Balanced acc.
Baseline: EfficientNet-B4, image-only	0.8100	65.0%	85.2%	75.1%
3-Model Ensemble + Late Fusion	0.8600	78.0%	81.5%	79.8%
Ours: 3-Model FiLM Ensemble + Prog. Resizing	0.9095	87.02%	79.15%	83.1%

Compared with the image-only baseline, our system improves sensitivity by +22.02 percentage points and ROC-AUC by +9.95 points; compared with late fusion, ROC-AUC improves by +4.95 points. Figure 5 shows the resulting sensitivity–specificity trade-off.

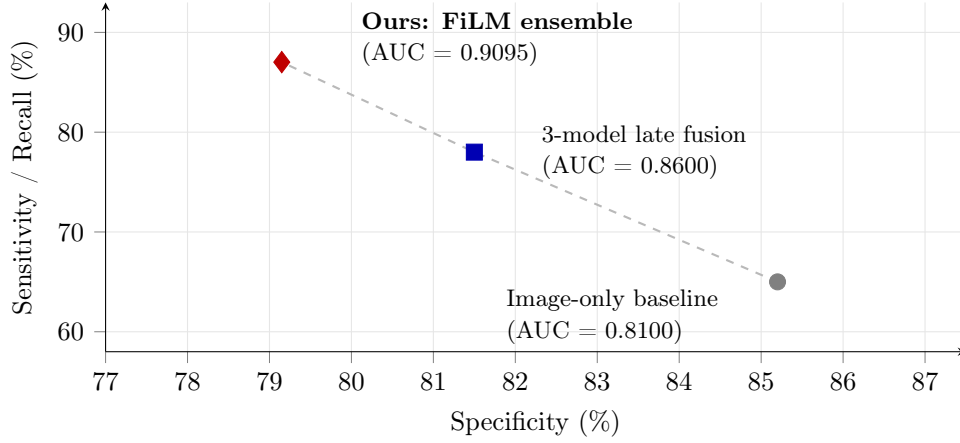


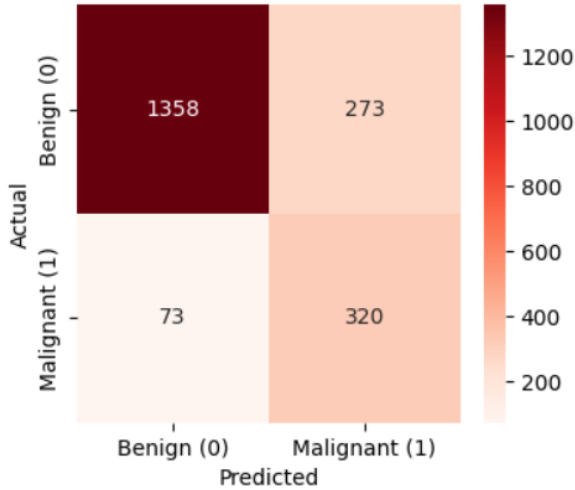
Figure 6: Sensitivity-specificity trade-off of the three configurations. The small specificity drop (81.5% $\rightarrow$ 79.15%) is justified by the +9 pp sensitivity gain.

Phase 1, Unimodal Baseline. Training EfficientNet-B4 on images alone (metadata discarded) yielded a ROC-AUC of 0.81. Critically, sensitivity was only 65%, meaning that one in three malignant lesions was missed by the image-only system. This confirms that pixel-level dermoscopic information alone is insufficient for reliable melanoma screening.

Phase 2, Late Fusion Ensemble. Reintroducing metadata via concatenation and ensembling three backbones raised the ROC-AUC to 0.86 (+5 points) and sensitivity to 78%. However, the improvement plateaued because the CNN feature extractors remained agnostic to clinical context during convolution.

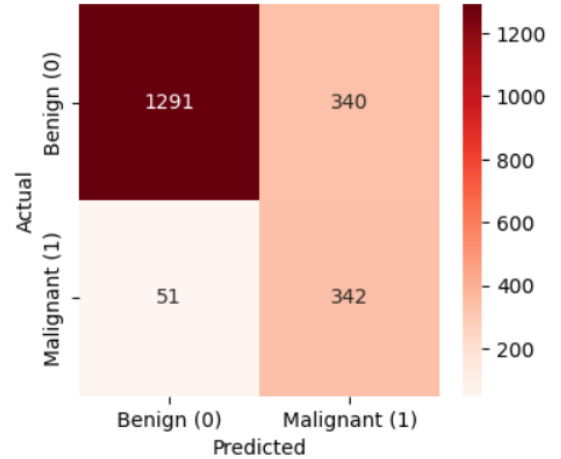
Phase 3, FiLM Ensemble (Ours). Replacing late fusion with FiLM conditioning produced the decisive breakthrough: ROC-AUC reached 0.9095 and sensitivity reached 87.02%. The progressive resizing strategy contributed additional gains by enabling model exposure to high-resolution (512 $\times$ 512) cellular micro-detail during Phase 2 fine-tuning.

Confusion Matrix — Default threshold (0.5)



(a) Default threshold (0.50)

Confusion Matrix — Optimal threshold (Youden)



(b) Optimal threshold (0.461)

Figure 7: Confusion Matrices comparing the default decision threshold against the Youden-optimised threshold. Lowering the threshold significantly reduces the number of false negatives (missed malignant cases) from 73 to 51, a critical improvement for clinical screening safety.

4.2 Individual Model Performance

Table 7: Individual FiLM model performance (before ensembling).

Model	ROC-AUC	Sensitivity	Specificity
EfficientNet-B4 (FiLM-conditioned)	0.882	83.1%	78.4%
ResNet-50 (FiLM-conditioned)	0.871	80.9%	77.2%
DenseNet-121 (FiLM-conditioned)	0.879	82.5%	79.0%
Ensemble (weighted average)	0.9095	87.02%	79.15%

The ensemble outperforms all individual models, confirming that the three backbones extract complementary, non-redundant feature representations.

4.3 Effect of Test-Time Augmentation

Table 8: Effect of TTA on FiLM ensemble performance.

Inference mode	ROC-AUC	Sensitivity	Specificity
Single pass (no TTA)	0.895	84.7%	78.8%
TTA (5 views)	0.9095	87.02%	79.15%
Δ improvement	+1.45	+2.32 pp	+0.35 pp

TTA provides a consistent 1.45-point AUC improvement by averaging out prediction variance introduced by the geometric orientation of the lesion in the image.

4.4 Clinical Deployment

The complete pipeline is deployed as an interactive **Streamlit** web application. A clinician uploads a dermoscopic image, provides the patient’s age, sex, and lesion localisation, and receives:

1. the original image alongside the hair-removed version (DullRazor output);
2. the ensemble’s malignancy probability with a clinical verdict (High Risk / Low Risk);
3. individual probability scores from each of the three FiLM models;
4. a visual risk gauge indicating the distance from the decision threshold.

The system processes a single case in under 5 seconds on CPU, making it suitable for resource-constrained clinical environments.

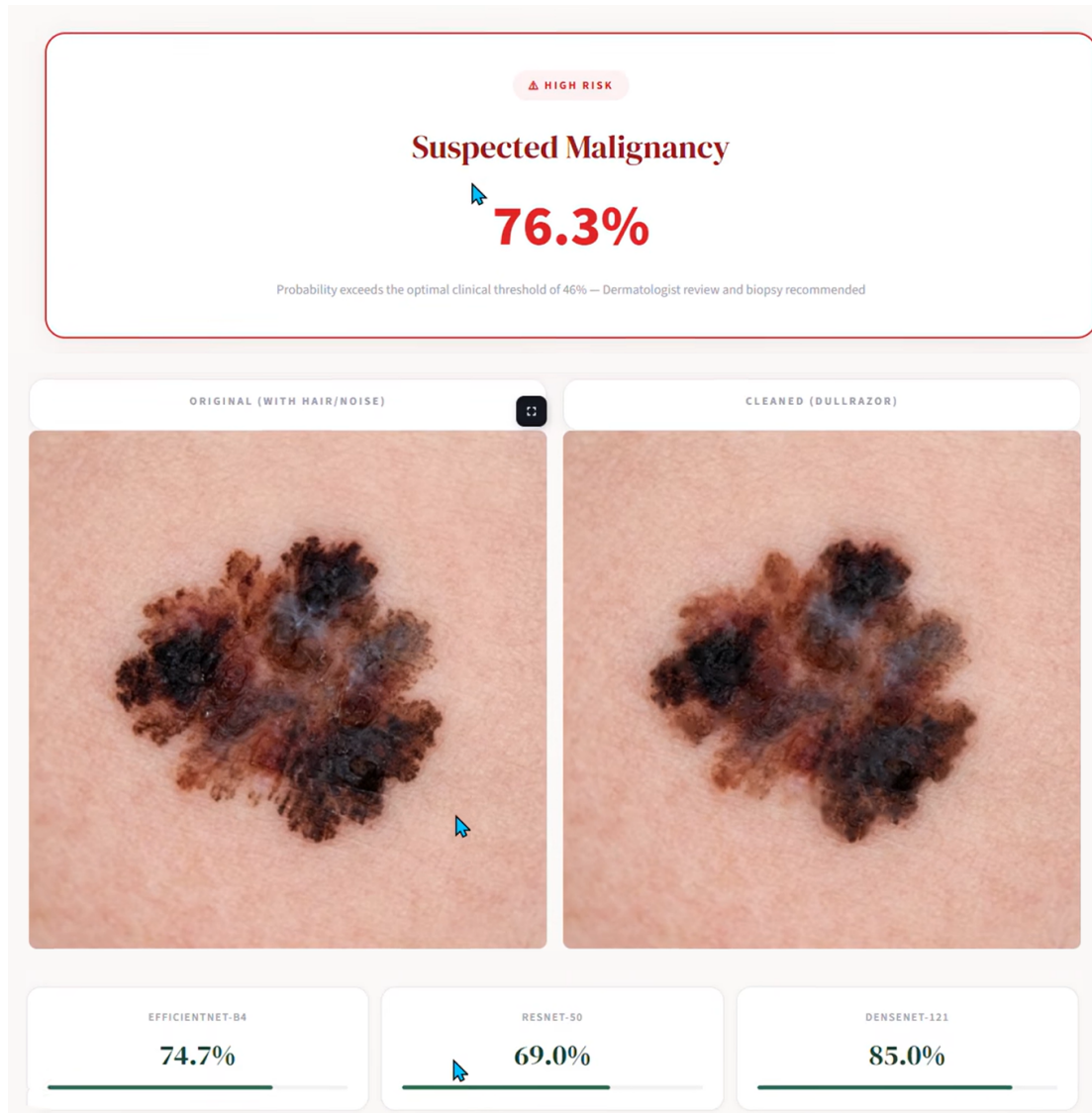


Figure 8: Interactive clinical deployment dashboard displaying the original and preprocessed images, alongside the ensemble’s final malignancy probability and the individual predictions from the three baseline models (the jury).

5. Discussion

5.1 FiLM versus Late Fusion: Why Architecture Matters

The ablation results demonstrate that **how** metadata is integrated matters significantly more than **whether** it is integrated. Late fusion improved performance (+5 AUC points), but FiLM’s ability to condition the feature extraction process itself, rather than merely appending metadata at the decision boundary, yielded an additional 4.95-point AUC improvement and a 9-point sensitivity gain.

The clinical interpretation is compelling: FiLM essentially allows the network to “know” that it is examining the back of a 70-year-old male before it begins analysing the lesion. The network can then selectively amplify channels sensitive to deep nodular morphology (associated with older males) and suppress channels attuned to flat, spreading patterns (more common in younger patients).

5.2 The Sensitivity–Specificity Trade-off

A notable observation in Table 6 is the slight decrease in specificity ($81.5\% \rightarrow 79.15\%$) from Phase 2 to Phase 3, accompanied by a substantial sensitivity increase ($78\% \rightarrow 87.02\%$). This trade-off is clinically desirable in a screening context. The cost asymmetry is stark:

- A **false negative** (missed melanoma) may result in delayed diagnosis, disease progression to an inoperable stage, and patient death.
- A **false positive** (benign lesion flagged for biopsy) results in a minor surgical procedure, temporary patient anxiety, and healthcare cost, all of which are manageable.

The Youden-optimised threshold (0.4607, below the conventional 0.5) explicitly encodes this clinical priority by shifting the decision boundary toward sensitivity.

5.3 Future Research Directions

1. **Cross-dataset generalisation:** validation on PH2 [17], Derm7pt [18], and ISIC 2018/2019 for distribution-shift robustness.
2. **Attention-based explainability:** integration of Class Activation Mapping (CAM) variants or SHAP-based metadata attribution to provide per-feature importance scores alongside spatial heatmaps.
3. **Skin phototype stratification:** separate validation sub-analyses stratified by Fitzpatrick skin type to assess and address demographic performance disparities.
4. **Longitudinal multimodal integration:** incorporation of patient visit history and lesion evolution data to enable temporal melanoma risk modelling.
5. **Federated learning:** privacy-preserving distributed training across hospital networks without centralising patient data.

6. Conclusion

We presented Derascope AI, a multimodal melanoma detection system that bridges the gap between pure computer vision and clinical dermatological reasoning through Feature-wise Linear Modulation. By conditioning three complementary CNN backbones on structured patient metadata at the feature level, rather than appending metadata at the decision boundary, our system fundamentally changes what the network sees based on who the patient is.

Our three-model FiLM ensemble, trained with progressive resizing and Focal Loss on the ISIC 2020 dataset, achieves a ROC-AUC of 0.9095 with a sensitivity of 87.02%, a 22-percentage-point improvement over a comparable image-only baseline. The system is deployed as a real-time Streamlit clinical dashboard and is openly available for research use.

The core contribution of this work is a demonstration of principle: in medical AI, fidelity to the clinical reasoning process, including all information a physician would use, produces measurably better outcomes than treating diagnosis as a pure vision task.

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